

Cross-National Differences in Predictors of Compliance with COVID-19

Protective Behaviors Before and After Mask Mandates

1 Introduction

Protective health behaviors such as mask wearing, social distancing, hand hygiene, and self-isolation were essential during the COVID-19 pandemic. However, compliance with these behaviors depended not only on government rules, but also on individual perceptions, trust, demographic characteristics, and social circumstances. Understanding these factors remains important because future public health responses will depend not only on introducing protective measures, but also on whether people in different national contexts are willing to follow them (Roma et al. 2020).

The aim of my project is to examine whether the predictors of compliance with COVID-19 protective behaviors differ across countries before and after mask mandates. The core research question is: do the factors associated with compliance change across policy periods, and do these changes differ between countries? Rather than estimating the causal effect of mandates, this project focuses on comparing how the predictive structure of compliance varies over time and across national settings.

2 Background

Existing research has identified a range of factors associated with compliance with COVID-19 protective behaviors, including demographic characteristics, risk perception, trust in government, and broader social and cultural influences (Galasso et al. 2020). Previous studies have also shown that protective behaviors often cluster together rather than occur in isolation, suggesting that compliance should be understood as a broader behavioral pattern rather than a single action. However, while much of this work has examined whether mandates increase compliance overall, much less attention has been paid to whether the predictors of compliance themselves change after mandates are introduced, and especially whether such changes differ across countries, regions, and cultural contexts. This leaves an important gap in understanding whether the drivers of compliance are stable across settings or shaped by different national and policy environments.

Ryan et al. (2025) addressed this issue in the Australian context by examining protective health behaviors before and after face mask mandates. Their findings suggest that the factors associated with compliance are not constant across policy periods: some predictors remained important throughout, whereas others became influential after mandates were introduced. The study therefore shows that policy context may reshape

the predictive structure of compliance, rather than simply change overall levels of adherence. At the same time, because the analysis focuses on Australia, it remains unclear whether similar patterns can be observed in other countries with different institutional, cultural, and policy settings. Building on this gap, the present project extends the analysis to a cross-national setting. This also motivates the use of harmonized cross-country data and comparative modelling to examine whether the predictors of compliance vary across both countries and mandate periods.

3 Methods

3.1 Data

For my project, I will use policy data from the Oxford COVID-19 Government Response Tracker (OxCGRT) to identify changes in COVID-19 mandates across countries. These policy records will be matched by country and date with behavioral survey data from the Imperial College London/YouGov COVID-19 Behaviors Tracker, which measures compliance with protective health behaviors. The main policy variable will be facial covering requirements, which will be used to distinguish pre-mandate and post-mandate periods (Hale et al. 2021).

3.2 Data preparation

Because this is a cross-national study, the first step I took is to organize the country-level policy records into a common country-date panel and standardize variable formats across datasets. This includes checking country names, date formats, missing values, and the consistency of policy indicators before linking them to behavioral data. Preliminary inspection of the policy data suggests that mandate periods can be identified consistently for a subset of countries with sufficient temporal coverage, which supports the feasibility of the proposed comparison.

3.3 Analytical approach

The analysis will proceed in two stages. First, I will use logistic regression as a baseline model to estimate how each predictor is associated with compliance. This provides an interpretable starting point and allows me to examine whether variables such as trust in government, perceived risk, or demographic characteristics are positively or negatively associated with protective behaviors in different policy periods.

Second, I will apply tree-based models, particularly random forests and XGBoost, to capture non-linear relationships and interactions that may not be identified by logistic regression alone. These models are appropriate because the relationship between compliance and its predictors is unlikely to be identical across countries or across pre-

mandate and post-mandate periods. Models will be estimated separately for pre-mandate and post-mandate periods, and the results will then be compared across countries. Where feasible, I will also estimate a pooled model with country effects to account for broader national differences while still examining individual-level predictors. Model performance will be assessed using standard classification measures such as accuracy, F1 score, and AUC. Comparing predictor importance across models, countries, and policy periods will allow the project to assess whether the predictive structure of compliance remains stable or changes across settings.

3.4 Initial plan

For my project, I will begin with a small subset of countries to test data quality, variable comparability, and the feasibility of defining common policy periods. If this stage is successful, the analysis will be extended to a wider cross-national sample. If harmonization proves too difficult, the project will be narrowed to a smaller set of countries or to a more consistent behavioral outcome, so that the study remains feasible within the available time and data constraints.

4 References

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